# BCSE498J- Project - II

## A HYBRID GRAPH-VECTOR RAG SYSTEM FOR FINANCIAL ANALYSIS

*Submitted in partial fulfilment of the requirements for the degree of*

## Bachelor of Technology

*in*

## Computer Science and Engineering (Bioinformatics)

*by*

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Under the Supervision of

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(Deemed to be University under section 3 of UGC Act, 1956)

**April, 2026**

# DECLARATION

I hereby declare that the project entitled "A HYBRID GRAPH-VECTOR RAG SYSTEM FOR FINANCIAL ANALYSIS" submitted by me, for the award of the degree of Bachelor of Technology in Computer Science and Engineering to VIT is a record of bonafide work carried out by me under the supervision of Prof. Meenakshi S P, School of Computer Science and Engineering.

I further declare that the work reported in this project report has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: Vellore **ADARSH (22BCB0096)**

Date :

# CERTIFICATE

This is to certify that the project entitled "A HYBRID GRAPH-VECTOR RAG SYSTEM FOR FINANCIAL ANALYSIS" submitted by ADARSH (22BCB0096), School of Computer Science and Engineering, VIT, for the award of the degree of Bachelor of Technology in Computer Science and Engineering (Bioinformatics), is a record of bonafide work carried out by the student under my supervision during Winter Semester 2025-2026, as per the VIT code of academic and research ethics.

The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university. The project fulfills the requirements and regulations of the University and in my opinion meets the necessary standards for submission.

Place: Vellore **Signature of the Guide**

Date :

**Internal Examiner** **External Examiner**

**Head of the Department**

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# TABLE OF CONTENTS

# EXECUTIVE SUMMARY

Analysts are drowning in unstructured financial documents like 10-Ks and earnings transcripts. I realized standard Retrieval-Augmented Generation (RAG) pipelines just can't handle this data. They rely entirely on vector similarity search. This breaks down completely when you ask a multi-hop relational question. I noticed the system kept losing the hierarchical document structure when I chunked the text into smaller pieces. It couldn't reconstruct causal chains between entities at all.

I built a Hybrid Graph-Vector RAG System to fix this exact problem. I parsed the raw financial texts and fed them through a dual-indexing pipeline. I used a FAISS vector store to catch the general semantic meaning. I struggled initially with extracting the hard facts, so I ran the text through LangChain's LLMGraphTransformer to build a Neo4j knowledge graph. This graph maps the entities, relationships, and time dependencies. I didn't want to use a slow LLM to decide where queries should go. I trained a Machine Learning Intent Router using TF-IDF and a Linear Support Vector Classifier (LinearSVC). It categorizes the incoming questions and sends them to the right database in under a millisecond. It is incredibly fast.

I tested the entire setup against a dataset of actual SEC 10-K filings. The hybrid strategy clearly beat the single-database baselines on multi-hop reasoning tasks. I recorded massive improvements in answer faithfulness and entity-level precision. I also managed to keep the overall query response latency very low. The architecture provides a reproducible framework for anyone building domain-specific financial tools. It actually works in practice.

# LIST OF TABLES

# LIST OF FIGURES

# LIST OF ABBREVIATIONS

| | |
|---|---|
| API | Application Programming Interface |
| BERT | Bidirectional Encoder Representations from Transformers |
| CPU | Central Processing Unit |
| EDGAR | Electronic Data Gathering, Analysis, and Retrieval |
| FAISS | Facebook AI Similarity Search |
| GNN | Graph Neural Network |
| GPU | Graphics Processing Unit |
| KGQA | Knowledge Graph Question Answering |
| LLM | Large Language Model |
| ML | Machine Learning |
| NER | Named Entity Recognition |
| PDF | Portable Document Format |
| RAG | Retrieval-Augmented Generation |
| RoBERTa | Robustly Optimized BERT Pretraining Approach |
| SEC | Securities and Exchange Commission |
| SVC | Support Vector Classifier |
| SVM | Support Vector Machine |
| TF-IDF | Term Frequency-Inverse Document Frequency |

# SYMBOLS AND NOTATIONS

| | |
|---|---|
| $\pm$ | Plus-Minus (Standard Deviation / Variance) |
| $+$ | Plus (Integration / Positive Margin) |
| $<$ | Less Than (Mathematical Threshold) |
| $\times$ | Times (Scale Multiplier) |
| $\%$ | Percentage (Statistical Metric) |
| $\rightarrow$ | Right Arrow (Prediction Mapping) |
| $*$ | Asterisk (Footnote Marker) |
| / | Forward Slash (Division Operator) |
| `()-[]->()` | Cypher Pattern (Graph Database Relationship) |

# CHAPTER 1

# INTRODUCTION

## 1.1 BACKGROUND

The financial world creates and uses data at a level that most other industries can't even imagine. I looked through a single SEC Form 10-K filing and it was a total monster, spanning hundreds of pages. These reports are packed with a messy mix of hard numbers, management talk, and tiny footnotes that hide crucial details. I realized that analysts are basically fighting a losing battle here. The sheer volume and the way the data is structured makes it impossible for a human to review everything manually without missing something important. It is just too much to handle.

Large Language Models and Retrieval Augmented Generation (RAG) seemed like the perfect way to automate these heavy financial tasks. RAG fixes the problem where an LLM only knows what it learned during its initial training by letting it look at outside documents on the fly. I spent a lot of time looking into the standard RAG pipeline where you turn a document collection into a vector database using an embedding model. When a user asks something, the system just pulls the top few chunks that look similar and hands them to the LLM to write an answer. It sounds simple enough. But I hit a snag when the PDFs were too messy for the standard parser.

This setup works great if you only need to find a single fact inside one document. It does its job well there. However, financial data doesn't fit into this neat little box at all. I found that financial reasoning depends on tracking relationships and time, which requires jumping across several different pages. You can't answer a question about how a 2022 goodwill impairment changed the next year's cash flows by just grabbing one paragraph. The system has to find the entity, follow the timeline, and piece together data from totally different sections. A basic search just misses the connection. It lacks the logic.

Vector similarity search—the core of standard RAG—simply cannot handle this kind of traversal. I realized that embedding-based retrieval just squashes a chunk of text into a fixed vector, which kills the actual relationship structure in the process. Close points in the vector space might mean the topics are related, but it doesn't mean there is

a causal link. Because of this, the system either gives a wrong answer based on a lucky word match or just fails to find anything. I saw this happen constantly during my early tests. The retrieval just breaks.

Knowledge graphs give us a completely different way to look at data. I found that by turning a document into a directed graph of entities and links, I could track the specific relationships that vector stores just ignore. I ran Cypher path queries over a Neo4j graph to navigate these multi-hop chains with actual precision. It finds things that a standard search completely misses. Getting the Cypher syntax right for these complex hops was a total pain, though. Graph-only systems also break if a user asks a question using words that don't match the graph's labels exactly. They are too rigid.

I decided to combine graph-based symbolic retrieval with vector-based semantic search into one hybrid system. This is the main focus of my project. I used the LLMGraphTransformer from LangChain to automate the graph construction so I didn't have to build the whole ontology by hand. I also added a simple, fast query router using TF-IDF and a Linear Support Vector Classifier. This tells the system which backend to use without the lag you get from using an LLM for every decision. It keeps things fast and reliable. My tests showed it stays way more consistent this way.

## 1.2 MOTIVATION

The motivation for this project emerges from three converging observations: one empirical, one architectural, and one operational.

### 1.2.1 The Empirical Observation: Multi-Hop Failure in Financial RAG

Standard RAG systems just don't work when you throw real financial questions at them. I found that when I used datasets like FinQA or ConvFinQA, the accuracy dropped off a cliff because the system couldn't connect info from different sections. The vanilla vector-RAG fails at things like tracking a timeline of events or figuring out which company a pronoun is actually referring to. It is frustrating to watch. It simply can't link the dots.

I've pinpointed the main issue as a total mismatch in how we represent the data. Vector embeddings focus on how words look next to each other, but they don't understand how financial facts actually relate. They don't know that a subsidiary's money is

part of a bigger segment total or that a new earnings report replaces an old one. These links are what actually matter in finance. Vector search just isn't built to find them. It misses the point entirely

### 1.2.2 The Architectural Observation: The False Dichotomy of Retrieval Paradigms

When I read through the past research, I noticed everyone treats vector search and graph search like they have to compete instead of working together. It makes no sense. I tested a bunch of these "specialist" systems and they instantly break the second you ask them something slightly outside their lane. Real financial questions are messy. Some questions just need a fuzzy text match, like asking about a company's general strategy across fifty pages of management spin. Other questions need hard numbers and exact links, like mapping out which subsidiaries generated what exact revenue in a specific footnote. You need exact graph math for that.

I built this hybrid setup to make a system that actually survives a full range of real-world financial questions without failing. No single database handles everything. I knew I had to build a routing engine that looks at a question's structure and decides exactly which backend gets the job. Writing the classification logic to actually sort these questions without adding massive latency was surprisingly hard. The router fixes the coverage gap.

### 1.2.3 The Operational Observation: The Latency Cost of LLM-Based Routing

I looked at how other hybrid systems handle query routing and saw everyone just prompts an LLM to classify the question. It technically works. But relying on a massive model to make every single routing decision adds a ridiculous latency cost, sometimes taking seconds just to figure out where to send the query before the search even starts. I cannot accept that kind of lag. Financial analysts expect the dashboard to snap back instantly. Waiting three seconds just for a basic routing decision ruins the whole system.

I realized that sorting queries is actually a pretty simple classification problem. You only need to look for specific triggers like relational keywords, exact entity names, or time-based comparisons. Old-school text classification handles this perfectly. I built the router using basic TF-IDF paired with a linear Support Vector Classifier. Tuning the SVC hyperparameters was incredibly tedious because the financial vocabulary is so

weird, but once dialed in, it runs in less than a millisecond on my laptop. It completely kills the latency bottleneck. The accuracy stays high.

Swapping out a heavy LLM for a fast, lightweight math classifier became one of the biggest technical wins of my project. People keep trying to use LLMs as universal remote controls for every single step of a pipeline. It creates bloated software. I rejected that trend entirely. I picked the right tool for the job.

## 1.3 SCOPE OF THE PROJECT

We defined the project scope across four specific dimensions covering the document domain, system architecture, technical components, and what we actually built for this implementation. We drew clear boundaries.

### 1.3.1 Document Domain Scope

we built and tested the system specifically for unstructured English financial text, focusing almost entirely on public company SEC Form 10-K annual reports. The 10-K is a parsing nightmare. It crams risk factors, management discussion, raw financials, and obscure footnotes into a single massive wall of text. My PDF extraction script kept crashing on the nested tables. We designed the core logic so it can eventually handle 10-Q quarterly reports or earnings call transcripts down the line. The methodologies are designed to generalise to similar financial document types, including 10-Q quarterly reports, earnings call transcripts, and analyst research reports, though those extensions are considered future work.

### 1.3.2 Architectural Scope

We built the end-to-end pipeline using five specific functional stages. It covers everything. And these five functional stages are:

First, I handle document ingestion and parsing. I wrote a script to load the raw 10-K PDFs, split the sections, and strip out the junk formatting. The parser kept reading table borders as actual text. I had to write custom regex just to fix it.

Next is the dual indexing phase where I build two databases at the exact same time. I spin up a FAISS vector index using sentence-transformer embeddings right alongside a

Neo4j knowledge graph. I fed the parsed text into LangChain's LLMGraphTransformer to extract the typed entity-relationship triples. It takes forever to run.

Then comes the ML-based intent routing. I trained a TF-IDF and LinearSVC model to look at the user's question and instantly decide whether to hit the vector search, the graph, or both. Fine-tuning the SVC weights took three days. It routes everything perfectly now.

After that, I run the actual retrieval and fusion. The system executes the chosen search method and merges the results if the router triggers the hybrid path. I wrote a strict protocol to stitch the hard facts from the graph together with the fuzzy text from the vectors. The data formats clashed constantly. I forced them to align.

Finally, we hit answer generation. I pass all that combined context to an LLM via an API so it can synthesize the final answer. I designed the code to track the source links all the way through the pipeline. You always know where the data originated.

### 1.3.3 Technical Component Scope

The system relies on a set of carefully chosen technologies: LangChain for orchestration, Neo4j as the graph database, FAISS for vector similarity search, and the LLMGraphTransformer for automated graph construction. We used Scikit-learn for developing the intent classification model, and the entire implementation is written in Python. A detailed justification for these choices is provided in Chapter 3.

### 1.3.4 Exclusions from Scope

We had to cut some of the planned features just to get a working prototype running. Like, the system does not include real-time data ingestion from financial APIs or streaming sources, relying instead on static documents. We avoided training a custom financial language model and a general-purpose LLM is used for answer generation to keep the design modular.

Also, our project does not address production-level concerns such as user authentication, multi-tenancy, or deployment infrastructure. The main goal was to develop a research prototype rather than a fully operational product.

We did not intend to create a complete financial intelligence platform. Our primary goal was demonstrating that the combined graph-based and vector-based retrieval,

guided by an ML-based intent routing mechanism, can improve performance on complex financial question-answering tasks. The outcome is a reproducible framework that supports further exploration in this area.

# CHAPTER 2

# PROJECT DESCRIPTION AND GOALS

## 2.1 LITERATURE REVIEW

The world of document retrieval and question-answering has shifted through several big architectural changes over the last twenty years. I spent a lot of time looking at this evolution to figure out where my project actually fits in the research world. It helps to look at the progress across four main areas: old-school search, neural semantic search, RAG, and how knowledge graphs work in big companies.

### 2.1.1 Classical Information Retrieval and Its Limitations in Financial Contexts

Early search systems for big companies used basic word-counting methods like TF-IDF and the BM25 ranking function. These algorithms basically just treat documents as a pile of keywords and rank them based on how many words overlap. I found that while BM25 was the industry standard for decades in legal and finance, it is actually pretty limited. It just looks at the surface text. It doesn't understand the actual meaning of the words.

I realized that lexical search fails the moment you use a different word for the same thing. If I search for ”revenue shortfall,” the system will totally miss documents talking about ”sales underperformance.” This is a massive problem in finance where ”net income,” ”profit after tax,” and ”bottom-line earnings” all mean the same thing but look different to a computer. My tests showed that BM25 starts to fall apart as soon as the questions get complex. The vocabulary mismatch is just too much.

These old-school methods also can't see how financial entities actually link together. A BM25 index just sees a document as a bag of words with no order. I found that it has no way to know that ”subsidiary revenue” belongs under ”consolidated group revenue.” It cannot see causal links between things like impairment losses and recoverable amounts. These links are the core of financial logic. Lexical systems are just blind to them.

### 2.1.2 The Neural Semantic Search Revolution

The shift to dense neural embeddings changed how we handle information retrieval forever. I looked into how Word2Vec and GloVe first proved that math could capture word meanings as geometric points in a vector space. It was a huge leap. Later on, models like ELMo and BERT moved this logic to full sentences and passages. These models encode actual context instead of just looking at the words on the page. They finally started to understand what the text actually meant.

Engineers call this approach dense retrieval or semantic search. Karpukhin et al. (2020) built the Dense Passage Retrieval framework to show that bi-encoders beat the old BM25 system on almost every benchmark. We can now search through millions of documents in milliseconds. I used the FAISS library to handle this at scale because it makes searching through billion-scale collections actually feasible. It uses hierarchical quantization to keep things fast. It really makes a difference.

Semantic search worked much better than old keyword methods in my financial tests. I found that domain-specific models like FinBERT performed way better after training on earnings transcripts and SEC filings. It finally understood the jargon. FinBERT embeddings found relevant passages for simple questions much faster than BM25 ever could. It was a game changer for single-hop queries.

However, the failures became obvious as soon as I looked at complex reasoning cases. Dense embeddings always lose information because they squash semantic meaning into a single messy vector. If a paragraph explains a company's debt covenant structure, the vector doesn't actually store the specific links between those covenants. It forgets which ratios they constrain or what happens during a breach. You cannot consistently find these tiny, specific details using just similarity search. Multi-hop queries fail because the relational structure just isn't there. Dense retrieval simply isn't enough on its own.

### 2.1.3 Retrieval-Augmented Generation: Foundations and Financial Applications

Lewis et al. (2020) created the Retrieval-Augmented Generation (RAG) framework, which became the standard template for almost all knowledge-heavy AI systems. RAG forces the LLM to generate answers based on external data it finds on the fly. This setup decouples actual facts from the model's internal parameters. We can now give the

AI up-to-date financial info without spending a fortune on retraining the whole model from scratch. It is a modular system. We can swap out the retrieval or generation parts whenever we want.

Early tests showed that RAG beats a basic LLM because it gives the model actual numbers to look at. AI is notoriously bad at remembering specific fine-grained data, so giving it the right passage to read helps a lot. But as I dug deeper into the research, I found that standard RAG pipelines have some massive flaws that make them unreliable for real work. My own experiments showed these same problems happening over and over. It was a pattern of failure.

**Chunking-Induced Context Loss:** We have to chop documents into small chunks before we can index them. Financial documents are massive, so the meaning often depends on connecting a line item in a financial statement to a footnote located fifty pages later. Fixed-size chunking or basic sentence boundaries systematically kill these cross-references. If my chunking script cuts that connection, the AI just guesses the missing info. It starts hallucinating. We watched it happen constantly with the fixed-size chunking I tried first. The context just disappears.

**Multi-Hop Retrieval Failure:** As we showed in Section 1.1, a single top-k retrieval pass cannot find the answer if the question requires connecting multiple steps across a document. The system just doesn't dig deep enough. Iterative methods like IRCoT try to fix this by mixing retrieval with chain-of-thought reasoning, but this makes the process painfully slow. It adds massive latency. Waiting for an LLM to think through five search steps makes the system feel laggy and broken. Plus, if the AI makes one logic error in the middle, the whole answer is trash. It's too fragile for what I need.

**Numerical Reasoning Instability:** Then there is the math problem. Financial questions always need you to calculate things like year-over-year changes or profit margins. Most RAG systems just feed the numbers to the LLM and hope it does the arithmetic correctly during its generation. It often fails. We know from plenty of tests that LLMs are terrible at math. They struggle with multi-step calculations or large, specific numbers. This instability comes from how the models predict the next token. You cannot fix a math problem by improving your retrieval.

**Entity Co-reference and Disambiguation Failure:** Financial documents are full of messy co-reference chains like pronouns, acronyms, and shorthand that point to spe-

cific companies or people. If the system doesn't explicitly resolve these links, the retrieved chunks end up with dangling references. The LLM then guesses wrong. It misresolves the entity and spits out a factual error in the final answer. The logic breaks.

### 2.1.4 Knowledge Graphs in Enterprise Financial Systems

Knowledge graphs are basically just maps made of points and lines that show how different things connect. They have been around in big companies way before this new AI era started. I found that banks and firms already use these for things like tracking supply chains or mapping out which company owns what. It is a very structured way to keep data organized.

Using these graphs to answer questions is called Knowledge Graph Question Answering, or KGQA. The idea is to turn a normal question into a formal code like Cypher or SPARQL. You then run that code against the graph to get a perfect, structured answer. I spent a lot of time looking at how this works because it offers a few huge wins over the standard vector search.

First, graphs actually understand relationships. Instead of guessing based on words, a graph has a direct link showing if one company owns another or has a specific legal debt. You can just query the link directly. My tests with Cypher showed that I could jump through multiple steps in a single go. I could write a query to find a company, its subsidiary, and its 2022 revenue all at once. It's deterministic, meaning it doesn't just "guess" the answer based on how words sound.

Graphs also handle time much better. I can add dates to the points and lines, which lets the system track how financial facts change over the years. A flat list of vectors just can't do that easily. Plus, everything is auditable. I can see exactly which link in the graph gave me the answer. This level of tracking is impossible with a standard vector search. It gives you a clear paper trail.

The big problem has always been building the graph in the first place. Doing it by hand is way too expensive and slow. I found that old-school rule-based systems were also a pain because they broke every time a company changed its reporting style. It just didn't scale.

Everything changed with LLM-based extraction. I used the LLMGraphTransformer in LangChain to solve this bottleneck. It uses the AI's understanding of language to pull

out the entities and relationships automatically. It has much better coverage and didn't require me to write thousands of lines of manual rules. This tool is really what made my whole project possible. I couldn't have built the graph without it.

### 2.1.5 Zero-Shot LLM Cypher Generation: The Instability Problem

A lot of recent research looks at using LLMs for direct natural language to Cypher (NL2Cypher) translation. We just prompt the LLM to write a valid Cypher query based on the user's question and a schema description. It looks like a great shortcut. This method seems to remove the need for a separate query translation part by letting the LLM handle everything. It simplifies the pipeline on paper.

In reality, testing zero-shot NL2Cypher reveals a massive instability that makes it useless for real financial systems. We found it completely unreliable. Our testing pinpointed four specific ways this approach fails:

**Schema Hallucination:** I noticed that the LLM constantly makes up its own labels for the graph even when I give it the exact schema. It basically ignores my setup and uses whatever financial terms it feels like using. This happens because the model mixes up its general training with the specific schema I built in Neo4j. It writes queries that look correct, but they fail instantly because the nodes don't exist. I spent hours debugging empty result sets. It kept hallucinating.

**Relationship Direction Errors:** Cypher is incredibly picky about which way the arrows point. If I write an "OWNS" relationship backwards, the whole meaning flips. My tests showed that the LLM gets these arrows wrong way too often, especially with complex stuff like consolidation hierarchies. I found that it struggles to distinguish who is the guarantor and who is the beneficiary. The logic just breaks. It gets messy fast.

**Query Complexity Collapse:** I found that the LLM completely falls apart when I ask it to write complex Cypher patterns for multi-hop queries. It can't handle variable-length paths or aggregation over subgraphs at all. Instead of doing the hard work, the model just simplifies the whole query. It gives me a basic string that looks okay but skips half the relational steps I actually need. The results end up totally wrong. It takes the easy way out.

**Non-Determinism and Brittleness:** I realized that the LLM is way too sensitive to how I word my questions. If I change a single word in a question, it might spit

out a completely different Cypher query that gives a different answer. This is a huge problem. You can't have a financial system that changes its mind based on a typo. The autoregressive nature of the model is just too unpredictable. It lacks the logic we need. Deterministic accuracy is a must.

All these failures convinced me that we just can't trust zero-shot Cypher generation for a real financial tool. It might work in a tiny research lab with a perfect schema, but it failed my stress tests. I decided to build the system differently to avoid this mess. We split the logic into two parts. First, an intent router decides if we even need the graph. Second, we use pre-written, validated Cypher templates instead of letting the LLM wing it on the fly. This kept my queries from breaking. It is much more stable.

## 2.2 RESEARCH GAP

The research review in Section 2.1 shows a massive gap in how we build these systems. I've identified three specific areas where the current tech just falls short, and that is exactly what I am trying to fix.

**The Retrieval Paradigm Dichotomy:**

I found that most researchers treat vector search and graph search like they are rivals. You either pick one or the other. This is a mistake. Systems built only on vectors fail when you ask them about relationships, and systems built only on graphs fail when the question is open-ended. There isn't a solid, hybrid architecture made specifically to handle the wide variety of questions you actually see in finance. I decided to stop choosing between them and just use both.

**The Routing Mechanism Vacuum:**

Most hybrid systems I looked at use either simple keywords or an expensive LLM to decide where to send a query. The keyword method is too fragile, and using an LLM adds a massive amount of lag that makes the system feel slow. I haven't found any prior work that tests a lightweight, trained ML classifier like TF-IDF paired with a LinearSVC for this job. It makes so much more sense to use a math-based classifier that runs in less than a millisecond. You don't need a giant generative model to solve a simple sorting problem. It's overkill.

**Automated Graph Construction for Financial Corpora:**

Building financial graphs from 10-K filings used to be a nightmare of manual work

or rigid rules. I realized that the scaling costs were just too high for anyone to actually do it well. While tools like the LLMGraphTransformer exist now, nobody has really benchmarked how good these graphs are when you build them from massive SEC reports. I want to see if the entity coverage and relationship accuracy are actually high enough to be useful. It's an area that hasn't been studied enough.

These three gaps are where my project lives. I've built a hybrid system that brings FAISS and Neo4j together under a fast ML router. By using the LLMGraphTransformer to handle the data, I can finally test if this architecture fixes the multi-hop reasoning failures that break standard RAG pipelines. My goal is to prove this approach actually works in a real-world setting.

### 2.3 OBJECTIVES

The following goals define exactly what I set out to build and test for this project:

**Objective 1:** I need to build a document ingestion pipeline that can parse and clean SEC 10-K filings. The goal is to prep this data for dual-indexing so it can sit in a FAISS vector store and a Neo4j knowledge graph at the same time.

**Objective 2:** I will use the LLMGraphTransformer from LangChain to automate the extraction of financial entity-relationship triples. I want to build a queryable Neo4j graph without having to spend weeks on manual ontology engineering.

**Objective 3:** I plan to train a TF-IDF and LinearSVC intent classification model. This router should look at a natural language question and send it to the right retrieval strategy—vector, graph, or hybrid-in less than a millisecond. My tests will focus on keeping the accuracy high despite the speed.

**Objective 4:** I have to design a hybrid fusion protocol to merge different types of data. It needs to take the hard facts from the graph and the semantic text from the vectors and stitch them into one clean context payload for the LLM.

**Objective 5:** I will run a head-to-head evaluation of my hybrid system against standard baselines like vector-only RAG and graph-only KGQA. I am using a curated list of financial questions that cover simple one-off searches, multi-hop links, and hard math problems.

**Objective 6:** I want to show clear, measurable wins in answer faithfulness and entity precision. I also want to prove that my ML routing mechanism significantly cuts down

on latency compared to using an LLM for the same job.

**Objective 7:** I intend to finish with a modular and well-documented system that actually works. It should serve as a solid foundation for anyone else who wants to research hybrid retrieval for specific domains later on.

## 2.4 PROBLEM STATEMENT

Standard RAG systems just don't work on messy financial reports. I found that they fail constantly when I ask questions that require jumping between different sections or tracking facts over a specific timeline. These are exactly the kinds of questions that actually matter for financial analysis. A basic search just can't keep up.

I've identified two main reasons for this. First, vector search is architecturally limited because it ignores how entities actually link together. Second, we lack a fast way to send a query to the right search engine. Most current hybrid systems are either too slow because they use an LLM for routing, or they rely on manual rules that just don't scale. I spent hours testing these bottlenecks. The lag was a huge issue.

**Specifically, I am tackling four big problems:**

**P1:** Vector RAG cannot solve multi-hop queries. Embedding-based search throws away the relationships between companies and numbers. I saw this lead to either total retrieval failure or the AI just making things up when it couldn't find a direct link. It's a mess.

**P2:** Graph systems have their own issues. I found that letting an LLM write Cypher queries on the fly is incredibly unstable. The model constantly hallucinates the schema or gets the relationship directions backwards. It collapses the moment the query gets complex. Purely symbolic search is just too rigid for real questions.

**P3:** No one has built a fast way to route these queries. Current systems either guess or use a heavy LLM that adds way too much latency. I want a math-based router that makes decisions in a fraction of a millisecond. We need efficiency, not more bloat.

**P4:** Building these graphs automatically is still a massive hurdle. I haven't seen a scalable pipeline that can take a raw 10-K and turn it into a high-quality graph without a ton of manual engineering. Getting the data out of those PDFs is a nightmare.

My research problem is simple: We have to build and test a hybrid system that actually works. It has to use the precision of a graph for the hard links and the flexibility

of vectors for everything else. I'm aiming for sub-millisecond routing and a graph that builds itself from raw text. We wanted to prove this beats the standard methods across the board. The goal is a system that finally handles the hard questions.

### 2.5 PROJECT PLAN

The project was executed across a structured timeline spanning one academic semester, organised into five phases with defined milestones and deliverables. The plan was designed to allow iterative refinement of components based on intermediate evaluation results.

**Phase 1 — Literature Survey and Architecture Design (Weeks 1–3)** I started by diving into the research on RAG, knowledge graphs, and financial NLP to see what was already out there. I had to define the hybrid architecture and decide exactly how the different parts would talk to each other. Picking the right technology stack early on was a big decision. I finished this phase with a solid design document and a clear reason for every tool I chose.

**Phase 2 — Data Pipeline and Dual Indexing (Weeks 4–7)** I spent these weeks building the actual pipeline to ingest 10-K documents. I integrated the LLMGraph-Transformer to handle the Neo4j graph construction and set up the FAISS vector index at the same time. Getting the graph to cover all the right entities was a struggle. By the end of week 7, I had both retrieval backends running and tested them on a small batch of documents to make sure they actually worked.

**Phase 3 — ML Intent Router Development (Weeks 8–10)** This was the most tedious part because I had to curate and label a dataset for the query classifier. I used TF-IDF for the feature engineering and then spent a lot of time on hyperparameter optimization for the LinearSVC model. I ran several latency benchmarks to make sure the routing stayed under a millisecond. I finally integrated the trained router into the main pipeline. It felt good to see it working.

**Phase 4 — System Integration and Hybrid Fusion (Weeks 11–13)** I finally hooked up the router to both retrieval backends. I had to write a custom protocol to fuse the results together whenever a hybrid query came through. Testing the whole pipeline on the full set of documents was a huge milestone. I spent a lot of late nights debugging weird errors where the data formats didn't match up. The end-to-end system was finally

operational.

**Phase 5 — Evaluation, Analysis, and Documentation (Weeks 14–16)** I ran the formal evaluation of my hybrid system against the basic vector and graph models. I used my curated list of financial questions to see if the hybrid approach actually won. Analyzing the numbers took a while, but it was worth it to see the improvements. I finished by writing up this report and getting my presentation ready. The final results are all documented now.

# CHAPTER 3

# TECHNICAL SPECIFICATION

## 3.1 REQUIREMENTS

### 3.1.1 Functional Requirements

I broke down the functional requirements to list exactly how the system needs to behave to meet my goals. Each part of the pipeline has its own set of rules.

**Document Ingestion:** The system has to take in SEC 10-K PDFs as the main input. I designed it to pull out all the raw text while keeping section headers and page numbers intact as much as I could. It also has to handle batch processing so I can feed it multiple documents at once without the script hanging. The parser was a nightmare to configure for multi-page tables. It needs to keep numbers exactly as they appear.

**Text Preprocessing and Segmentation:** I need the system to chop up the text into chunks that actually make sense for the databases. These chunks have to follow sentence or paragraph lines instead of just cutting off at a random character count. I've seen standard RAG systems break sentences in half, and it completely ruins the search quality. It must also tag every single chunk with metadata like the file name and page number. I made sure these attributes stay attached through the whole indexing process. This makes tracking the source much easier later on.

**Vector Index Construction:** The system has to turn every document chunk into a dense vector embedding using a sentence-transformer model. I picked a FAISS vector store to index these because it is incredibly fast for top-k searches. I made sure the index can be saved to the disk and reloaded later. Re-indexing every time. I started a new session was a huge waste of time. I fixed that early on. The search depth stays configurable so I can tweak the results.

**Knowledge Graph Construction:** I set up the pipeline to feed document chunks into the LLMGraphTransformer to pull out entity-relationship triples. These triples get stored as nodes and directed edges inside a Neo4j database. I had to write a deduplication step to make sure the same company doesn't show up as five different nodes just because of minor name variations. Every node and edge in the graph carries metadata that links it back to the original source text. This makes the data much easier to verify.

It keeps the graph clean and organized.

**Intent Classification and Routing:** The system takes in a natural language string and runs it through the TF-IDF and LinearSVC model I trained. The classifier has to sort the question into one of three buckets: VECTOR, GRAPH, or HYBRID. I kept the logic simple so this happens in less than a millisecond. It also logs the routing decision along with a confidence score. I needed this feature to debug why certain questions were being sent to the wrong backend during my tests. It works much faster than any LLM-based alternative.

**Vector Retrieval:** When the router picks the VECTOR path, the system embeds the question using the same transformer I used for the indexing. It then grabs the top-k most similar chunks from FAISS. I ranked these results by cosine similarity to make sure the most relevant text sits at the top. Everything comes back with its original metadata so I don't lose the page numbers. It is a very direct process.

**Graph Retrieval:** For GRAPH-routed questions, the system runs a parameterized Cypher query against Neo4j. I designed the code to pull entities from the user's question and use them as starting points for the graph traversal. The resulting subgraphs get turned into a structured text format so the LLM can actually read them. Writing the serialization logic was a real pain because the graph data is so nested. It has to be clean for the model to understand it.

**Hybrid Retrieval Fusion:** If a query hits the HYBRID class, the system kicks off both the vector and graph searches at the same time. I built a fusion protocol to stitch these two different data types together into one payload. It keeps the graph facts and the vector passages in their own labeled sections so the LLM doesn't get confused. Merging them without hitting token limits was a constant struggle during my tests. The unified context stays organized and readable.

**Answer Generation:** The system passes the final context payload and the original question to the Google Gemini 1.5 Pro API. I wrote a system prompt that forces the AI to stay in "financial analyst" mode so it doesn't wander off-topic. The final answer goes back to the user with clear source tags showing if the data came from a vector chunk or a graph node. I made sure it identifies the exact retrieval path used. This makes it easy to see if the router made the right call.

**API Endpoint Exposure:** I used FastAPI to turn the query interface into a RESTful

API. It accepts POST requests with a JSON body for the query and any extra settings I want to tweak. The system then spits out a structured JSON response with the answer, the routing choice, and all the sources. I also added latency metrics to the response so I could track how fast everything was running during my tests. Setting up the Pydantic models for the JSON validation was a bit of a chore. It keeps the data clean, though.

**Query Logging and Observability:** I made sure the system logs every single query and its routing decision to a persistent file. It records the retrieval results and the final answer so I can go back and analyze them later. I also track the timing for every stage, from the initial routing to the final generation. This helped me find exactly where the bottlenecks were happening. I found that the Gemini API call is usually the slowest part of the whole thing. Everything else is almost instant.

### 3.1.2 Non-Functional Requirements

I set these non-functional requirements to make sure the system actually performs under pressure, not just in theory. These limits kept my development focused on speed and reliability.

**Routing Latency:** The TF-IDF and LinearSVC classifier has to finish routing in under 5 milliseconds on my laptop. I measured this as the total wall-clock time from when the string enters the system to the final decision. This is the main reason I built this specific router. I wanted to prove that you don't need a heavy LLM to make basic logic choices. My tests showed it usually runs even faster than this 5ms limit.

**End-to-End Response Latency:** I am aiming for a total response time of under 10 seconds from the moment a user hits the API to the final answer. I don't count the Gemini API network lag here because that is out of my hands. I found that my local processing routing and searching takes up almost no time at all compared to the generation step. Keeping the local overhead low is the only way to make the system feel snappy. It keeps the UI from hanging.

**Retrieval Accuracy:** For retrieval quality, We set a baseline of 90% accuracy for the intent classifier on a labeled set of financial queries. We also set a goal for the vector search to hit a Recall@5 of 0.75 for simple, single-hop questions. This means the right answer should be in the top five results 75% of the time. Tuning the SVC to distinguish between ”semantic” and ”relational” questions was harder than I thought. I had to go

back and relabel my training data three times.

**Scalability:** I wanted both the FAISS index and the Neo4j graph to handle at least 50 full 10-K documents without a noticeable drop in performance. I set a limit that retrieval can't slow down by more than 20% even as the database grows from one document to fifty. It was a challenge to keep the graph traversal efficient as the number of nodes climbed. I had to optimize my Cypher queries to avoid full-graph scans.

**Modularity:** also put some effort into keeping the system modular. Each stage ingestion, indexing, routing, retrieval, and generation runs as its own component with defined inputs and outputs. Each one has a clear input and output interface so I can test them one by one. I made sure I could swap out the LLM at the end without touching the rest of the code. This made debugging much easier when the Gemini API was down. I could just plug in a local model and keep working.

**Reproducibility:** For the routing and retrieval stages, I wanted completely consistent behavior. If I feed the system the same question twice, it has to make the same routing decision and grab the same data every time. Since LLMs are inherently random, I added a temperature setting to the generation stage. Setting the temperature to zero helped me get consistent answers during my evaluation runs. It made debugging the final output much less of a headache.

**Fault Tolerance:** I built the system to handle failures without just crashing. If the graph search comes up empty because it couldn't find a specific entity, the code automatically falls back to a standard vector search. I made sure the API flags this fallback event in the response so I know exactly what happened. On top of that, the FastAPI layer handles runtime errors and returns structured JSON responses, which avoids dealing with raw stack traces during debugging.

**Security:** The system has to validate every incoming request against a strict schema before it even starts processing. I set up the Neo4j connection to require authenticated sessions so the database isn't just sitting open. All my API keys for Gemini stay tucked away in environment variables. I was careful to make sure these keys never show up in the logs or the API responses. It is a basic precaution, but I didn't want to leak my credentials during testing.

**Portability:** The system is packaged to run in a Docker container, which keeps the environment consistent across different machines. The system just needs network

access to hit the Neo4j instance and the Gemini API. I pinned every single version number in the requirements file to avoid the ”it works on my machine” problem. My first attempt at a Docker build failed because of a version conflict in the LangChain library. I fixed that by being very specific with the dependency list.

### 3.2 FEASIBILITY STUDY

I checked the maturity of all the tools I picked to make sure this project was actually possible to build. Each main part of the stack has already been proven in real-world apps and research papers.

**LangChain and LLMGraphTransformer:** LangChain is a solid orchestration framework with a huge community and plenty of documentation. I used the LLMGraphTransformer module because it is known for pulling triples out of general text. Adapting it to 10-K filings is a new twist, but the core idea of using an LLM to extract structured data is a solved problem. I found that as long as the prompt is clear, the extraction logic holds up. It saved me from writing a custom parser from scratch.

**FAISS:** Facebook AI Similarity Search is a battle-tested library that companies like Microsoft use for massive vector retrieval. It has great Python bindings and plugs directly into LangChain, which made my life much easier. I didn't have to worry about the math behind the similarity search because FAISS handles it so well. It is incredibly fast.

**Neo4j:** Neo4j is the most popular graph database out there. It uses the Cypher query language and has a very stable Python driver that I integrated into my pipeline. The property graph model it uses fits the entity links in financial reports perfectly. I found that even with complex nested relationships, Neo4j stays responsive. It was a reliable choice for the backend.

**Scikit-learn LinearSVC:** Linear Support Vector Machines are some of the most reliable classification algorithms out there. I picked the TF-IDF and LinearSVC combo because it's a classic, proven way to handle text classification without needing a massive server. It is incredibly lightweight. My tests showed it handles the high-dimensional feature space of financial terms much better than I expected. It just works.

**Google Gemini 1.5 Pro:** I chose Gemini 1.5 Pro because its huge context window up to a million tokens is a lifesaver for this project. Financial reports are long, and

sometimes my retrieved context payloads get pretty big. Older models would have choked on that much data. The API is stable and easy to hit through Google AI Studio. It handles the heavy lifting of the final answer generation while I focus on the retrieval logic.

**FastAPI:** This is a modern Python framework that I used for the API layer. It's built on Pydantic, so it handles the data validation and schema generation automatically. I found the asynchronous request handling really useful for keeping the system responsive while waiting on the databases. It is exactly what I needed to make the system accessible over HTTP. It's fast and didn't give me much trouble during setup.

Putting all these parts together is definitely possible. I haven't run into anything that requires a tool to do more than it was designed for. All the links between the parts like the LangChain wrappers and the Neo4j driver are well-documented and kept up to date by the developers. I'm confident the architecture is solid. The integration was surprisingly smooth once I got the versioning right.

## 3.3 SYSTEM SPECIFICATION

### 3.3.1 Hardware Specification

Table 3.1: Hardware Specification

| **Component** | **Specification** |
|---|---|
| Processor | Intel Core i7 / AMD Ryzen 7 (8-core, 3.5 GHz+) or equivalent |
| Memory | Minimum 16 GB RAM (32 GB recommended for large corpus indexing) |
| Storage | Minimum 50 GB SSD (for document corpus, FAISS index, and Neo4j data store) |
| GPU | Optional; CUDA-compatible GPU accelerates embedding computation |
| Network | Stable internet connection required for Gemini API access |

### 3.3.2 Software Specification

Table 3.2: Software Specification

| Component | Technology | Version | Role |
|---|---|---|---|
| Programming Language | Python | 3.10+ | Core implementation language |
| Orchestration Framework | LangChain | 0.2.x | Pipeline orchestration, LLMGraphTransformer, vector store integration |
| Graph Database | Neo4j | 5.x (Community Edition) | Knowledge graph storage and Cypher query execution |
| Vector Index | FAISS | 1.7.x | Dense vector indexing and approximate nearest-neighbour retrieval |
| ML Classification | Scikit-learn | 1.4.x | TF-IDF feature extraction and LinearSVC intent classification |
| LLM Generation & Extraction | Gemini 1.5 Pro API | (latest stable) | Knowledge graph triple extraction and answer generation |
| Embedding Model | sentence-transformers (all-MiniLM-L6-v2) | 2.x | Document and query dense embedding |
| API Framework | FastAPI | 0.110.x | RESTful API layer with request/response handling |
| PDF Processing | PyMuPDF (fitz) | 1.23.x | PDF text extraction with layout preservation |
| Graph Driver | neo4j-driver (Python) | 5.x | Authentication and Neo4j session management |
| Environment Management | python-dotenv | 1.0.x | Secure API key and configuration management |
| Data Validation | Pydantic | 2.x | Request/response schema validation within FastAPI |

# CHAPTER 4

# SYSTEM DESIGN

## 4.1 SYSTEM ARCHITECTURE

I organized the Hybrid Graph-Vector RAG System into a two-phase pipeline. These two modes are separate: the indexing phase, which happens offline whenever I update the documents, and the query phase, which runs in real-time whenever a user asks a question. Both phases use the FAISS store and Neo4j graph as a shared data plane, but they don't actually depend on each other for logic.

I made this separation on purpose for two main reasons. First, indexing is a massive resource hog especially the part where the LLM builds the graph so I can't have that happening while a user is waiting for an answer. Latency would be terrible. Second, this setup lets me add new documents to the corpus incrementally. I can update the databases without having to touch or restart the query pipeline at all. It makes the whole thing much more stable.

The architecture is built around seven logical parts that I've been working on. I'll get into how they actually talk to each other in Section 4.2, but here is the high-level breakdown:

1. **Document Ingestion Module —** This handles the messy work of loading and cleaning the raw PDFs from SEC filings.

2. **Dual Indexing Engine —** This is the ”brain” that manages the parallel builds for both FAISS and Neo4j.

3. **FAISS Vector Store —** My persistent index for quick similarity searches.

4. **Neo4j Knowledge Graph —** The database where I store all those financial entity nodes and relationship links.

5. **ML Intent Router —** The TF-IDF + LinearSVC model I trained to intercept queries and decide where they should go.

6. **Hybrid Retrieval Orchestrator —** This follows the router's orders to run the search and stitch together the context payload.

7. **Generation and Response Module —** This sends everything to Gemini 1.5 Pro and cleans up the final API response.

For the outside world, the system connects to SEC EDGAR to get the files, the Google Gemini API for the heavy AI work, and the FastAPI interface so users can actually send their questions. Keeping these external connections modular helped me a lot during the integration phase. If one API was acting up, it didn't break the local logic of the other modules.

## 4.2 DETAILED DESIGN

### 4.2.1 The Data Pipeline: From Raw PDF to Dual Index

The pipeline is designed to turn a messy, unstructured PDF into two very different structured databases: a FAISS vector index for fuzzy matching and a Neo4j property graph for hard logic. I spent a lot of time getting this transformation right because if the data goes in dirty, the answers come out wrong.

**Step 1: PDF Loading and Raw Text Extraction**

The process starts with a raw PDF usually a massive 10-K filing that can be over a hundred pages long. I chose to use PyMuPDF (the fitz library) for this. I tested simpler tools like pdfplumber, but they kept mangling the layout. PyMuPDF is much better at keeping track of where text actually sits on the page. This lets me use font sizes and positioning to figure out what is a section header and what is just a footnote. It's not perfect, but it's much more reliable.

The script pulls text page by page. For every page, I get the raw string and the coordinates for the text blocks. I then run a custom cleaning routine to strip out the "junk" like page numbers and document titles that repeat at the top of every sheet. I also had to write a fix for non-ASCII symbols, like those weird em-dashes and currency symbols that tend to break the vectorizers later on.

One thing I added was a "table flag." If a page has way more numbers than letters, the script marks it as tabular data. I found this is a lifesaver for downstream handling because tables usually need different processing than standard paragraphs. The final output of this step is just a clean list of page-level strings, each tagged with the filename and page number so I never lose the source.

**Step 2: Semantic Chunking**

I found that raw page-level text is just too bulky. It's too big for a search engine to be precise and way too long for the LLMGraphTransformer to handle without losing its mind. To fix this, I had to break the text into smaller, retrieval-ready pieces.

I used LangChain's RecursiveCharacterTextSplitter for this part. I set it to a chunk size of 1,000 characters with a 200-character overlap. The "recursive" part is the most important—it tries to split on double newlines first to keep paragraphs together. If that doesn't work, it tries periods to keep sentences whole. It only cuts mid-sentence if it absolutely has to. I hated seeing half-sentences in my early tests, and this approach mostly solved that.

The 200-character overlap was a very specific choice. Financial reports love to put a company name at the end of a page and its revenue at the start of the next. Without that overlap, the connection between the entity and the number gets cut in half. My tests showed that cosine similarity doesn't care about the slight duplication in the FAISS index, but the retrieval of these "boundary" facts got way better once I added the overlap.

Every chunk ends up as a LangChain Document object. I made sure each one carries a metadata dictionary with the filename, page number, and a sequential chunk id. I also tried to flag the section name, like "Risk Factors" or "MD&A," whenever the parser could figure it out. This gives me a massive list of Document objects that are finally ready to be indexed into the two backends.

**Step 3: FAISS Vector Index Construction**

Vector indexing happens in two main parts: calculating the embeddings and then actually building the index. I spent a lot of time testing different models before landing on this setup.

**Embedding Computation:** I ran every document chunk through the all-MiniLM-L6-v2 model. It turns the text into a 384-dimensional vector. I picked this specific model because it's a great middle ground—the quality is high, but it's fast enough to run on my laptop's CPU without a dedicated GPU. It churns through hundreds of chunks per second. I made sure to use the same model for the user queries later on. If the models don't match, the math behind the search fails completely because the vectors won't be in the same "space."

**Index Construction:** I pushed these vectors into a FAISS IndexFlatIP (Inner Product) index. Since I'm normalizing the vectors first, this is essentially calculating exact cosine similarity. For the size of my project—about a few dozen 10-K files—this exact search is fast enough and much better than using "approximate" shortcuts like HNSW or IVF. I didn't want to lose any accuracy just to save a few milliseconds on a search that was already nearly instant.

I then saved the FAISS index to the disk using `faiss.write_index()`. I also had to create a separate JSON file to map the internal FAISS IDs back to my Document metadata. Without this mapping, the index just returns a bunch of numbers with no way to tell which page or file they came from. It was a bit of extra work to keep these files in sync, but it's the only way to get proper source attribution for the final answers.

**Step 4: Knowledge Graph Construction via LLMGraphTransformer**

Building the graph is by far the most resource-heavy part of the pipeline and definitely the most experimental. I fed each document chunk into LangChain's LLMGraphTransformer, using Google Gemini 1.5 Pro as the engine. Since the graph construction is slow, I had to be very careful with how I structured the extraction to avoid wasting API credits on junk data.

The `LLMGraphTransformer` operates by prompting Gemini to identify named entities within the input text and the relationships between them. The extracted information is returned as a structured list of triples of the form `(subject, relationship, object)`.

The model is further constrained to assign a predefined semantic type to each entity and relationship. For example, entities are labelled as *Company* or *Subsidiary*, while relationships are classified as *OWNS* or *REPORTED_BY*.

It was observed that, without a strict predefined schema of allowed types, the model tends to generate a large number of inconsistent and semantically irrelevant labels. Enforcing a controlled vocabulary ensures consistency, improves graph quality, and prevents the proliferation of redundant or meaningless entity and relationship types.

To fix this, I defined a custom "allowed list" of entity and relationship types. This was a big design choice to stop "schema hallucination" right at the start. By forcing the graph to follow a strict financial schema during the building phase, I don't have to worry about the system trying to query categories that aren't actually there later on. It

keeps the database focused and clean.

Once I have the triples, I convert them into `GraphDocument` objects and push them into Neo4j using the `add_graph_documents()` method.

I used `MERGE` statements instead of simple `CREATE` statements for the Cypher queries. This is the only way to handle Entity Deduplication.

If "Apple Inc." is mentioned in ten different chunks, I want one node for the company, not ten separate ones.

I also wrote a normalization script to handle minor differences like "Inc." vs "Incorporated" or extra punctuation.

Without this, the graph would just be a fragmented mess of disconnected nodes.

The final result is a Neo4j graph with thousands of nodes and edges that actually map out how the financial entities are connected. It turns the raw text into a giant, queryable map of the company's business structure. It's a lot more powerful than just having a pile of text chunks.

### 4.2.2 The Query Pipeline: How the SVM Intent Router Intercepts and Directs Traffic

The query pipeline is activated for every incoming user query submitted through the FastAPI endpoint. It proceeds through five sequential stages: query reception and validation, intent classification and routing, retrieval execution, context fusion, and answer generation.

**Stage 1 Query Reception and Validation**

A user submits a POST request to the /query endpoint of the FastAPI application, with a JSON body.

FastAPI's Pydantic integration validates the request body against the defined QueryRequest schema, confirming that required fields are present and correctly typed. The validated query string is extracted and passed to the intent classification stage. A pipeline execution timestamp is recorded at this point for end-to-end latency measurement.

**Stage 2 Intent Classification and Routing (The SVM Intent Router)**

The ML Intent Router is the architectural centrepiece of the query pipeline. Its design and operation are described in full technical detail here.

**Router Architecture:** The router consists of a two-stage transformation: a TF-

IDF vectoriser followed by a trained LinearSVC classifier, both implemented in Scikit-learn and serialised to disk using joblib after training. Both components are loaded into memory at application startup and remain resident for the lifetime of the FastAPI process, eliminating any model loading latency from the per-query critical path.

**TF-IDF Feature Extraction:** The incoming query string is passed to the fitted `TfidfVectorizer`, which maps it to a sparse high-dimensional vector over the vocabulary established during training. The vectoriser is configured with unigram and bigram features (`ngram_range = (1, 2)`), a maximum vocabulary size of `10,000` features (`max_features = 10000`), and sublinear TF scaling (`sublinear_tf = True`).

Sublinear TF scaling — replacing raw term frequency $f$ with $1 + \log(f)$ — prevents high-frequency terms from dominating the feature vector, which is particularly important in financial text where terms such as "the," "company," and "fiscal" appear at very high frequencies across all query types.

**LinearSVC Classification:** The TF-IDF sparse vector is passed to the fitted LinearSVC classifier, which computes the dot product of the input vector with each of the three learned weight vectors (one per class: VECTOR, GRAPH, HYBRID) and assigns the query to the class with the highest decision function score. This dot product computation over a sparse vector is extraordinarily efficient completing in microseconds on a standard CPU because only the non-zero features of the TF-IDF vector participate in the computation.

**Routing Decision Logic:** The classifier sorts the user's intent into one of three distinct paths based on the linguistic structure of the query:

1. **VECTOR:** This path is for open-ended, thematic, or "vibe-based" questions. I found that these queries usually lack specific entity chains and focus more on qualitative descriptions. For example, if someone asks, "What are the main risk factors disclosed by the company regarding its international operations?", the vector store is the best place to find those descriptive paragraphs.

2. **GRAPH:** This is where the precision of the Neo4j backend shines. I trained the router to look for specific entity names and relational verbs like "guarantees" or "owns." An example would be: "Which subsidiaries are identified as guarantors of the parent company's senior unsecured notes?" This requires a direct hop through the graph, not just a keyword search.

3. **HYBRID:** This is the most complex category. It's for questions that need both the "why" (semantic) and the "what" (relational). For instance, "How does the company characterize the strategic rationale for its acquisition of [Entity X], and what financial metrics are reported for [Entity X]?" needs the vector store for the rationale and the graph for the specific metrics.

I made sure the system logs the final decision, the raw scores for all three classes, and the exact classification time in microseconds. It's been really helpful to see those logs during my evaluation phase to verify exactly how the router is "thinking."

**Stage 3 — Retrieval Execution**

I designed the retrieval stage to follow the router's decision exactly. Each path works as an independent, asynchronous function, which the orchestrator calls based on the classification.

**VECTOR Retrieval Path:** The system turns the query into a 384-dimensional vector using the same all-MiniLM-L6-v2 model from the indexing phase. I make sure this vector is L2-normalized before hitting the FAISS search() method. It pulls the $k$ nearest chunks based on cosine similarity. I then use the mapping file to swap those internal FAISS IDs for actual Document objects. I found that sorting these by similarity score in descending order gives the LLM the best context to work with.

**GRAPH Retrieval Path:** This part is a bit more complex. I apply Named Entity Recognition (NER) to the question to find starting points—like company names or specific metrics. Instead of letting an AI hallucinate the database code, I use parameterized Cypher templates. The system picks a template based on the question (like a single-hop lookup or a multi-entity path) and plugs in the extracted names. I execute this against Neo4j through an authenticated session.

I then serialize the results into a clear format:

[ENTITY: name, TYPE: label] –[RELATIONSHIP: type]–¿ [ENTITY: name, TYPE: label].

By using these templates rather than zero-shot LLM generation, I completely avoided the schema errors and "complexity collapse" I read about in earlier research. It makes the graph retrieval deterministic and much more reliable for a real-world project.

**HYBRID Retrieval Path:** If the router picks hybrid, the system just runs both the vector and graph paths. I used LangChain's asynchronous utilities to run these searches

in parallel. I didn't want to double the wait time by running them one after the other. It's a bit more complex to manage the threads, but it keeps the latency low enough that you don't even notice the extra work happening in the background.

**Stage 4 — Context Fusion and Assembly**

Once the retrieval is done, I have to stitch the data together so the LLM can actually make sense of it. The way I assemble the final "payload" depends entirely on which path the router chose.

1. For VECTOR-only queries: I take the document chunks and line them up in order of their similarity scores. I add a header at the top that tells the AI exactly where this text came from, including the source metadata like page numbers. It's a pretty standard approach that works well for qualitative questions.

2. For GRAPH-only queries: I take that serialized subgraph from Neo4j and wrap it in a structured block. I labeled it clearly with a header: [GRAPH RETRIEVAL RESULTS SOURCE: Neo4j Knowledge Graph]. I found that giving the AI a clear signal that this is "structured fact" data helps it stay grounded.

3. For HYBRID queries: This is where the fusion gets interesting. I structure the payload into two distinct, labeled sections. I explicitly tell the LLM that it is looking at two different types of information from different sources.

I wrote the system prompt to give the AI specific instructions on how to handle this mix. I told it to prioritize the graph facts for hard relational claims like who owns what and use the vector passages for the "fluff" or the qualitative explanations. This way, the model doesn't try to guess a relationship from a vague paragraph if the graph has the hard link right there. It makes the final answer feel much more analytical and precise.

**Stage 5 Answer Generation and Response Formatting**

I take the final context payload, the original query, and my custom system prompt and send them over to the Google Gemini 1.5 Pro API using the official Python SDK. I wrote the system prompt to be very strict—it tells the model to stay in "financial analyst" mode and ground every single word in the provided context. I explicitly forbade it from using its own training data to fill in the gaps. I also required it to cite exactly which sections it used when making a claim. This was the only way I could ensure the answers weren't just plausible-sounding hallucinations.

Once I get the text back from Gemini, the FastAPI handler kicks in to build a structured JSON response. I didn't want the API to just return a string of text; I wanted it to be a full diagnostic report for the user. The final JSON includes:

1. **answer:** The actual response from the AI.
2. **routing_decision:** Which path the LinearSVC chose (VECTOR, GRAPH, or HYBRID).
3. **classifier_scores:** The raw scores for all three classes so I can see how "sure" the router was.
4. **retrieval_sources:** A full list of metadata for the vector chunks and a summary of the graph nodes we hit.
5. **latency_breakdown:** The timing for every single stage in milliseconds.
6. **fallback_triggered:** A simple true/false flag so I know if the graph failed and we had to resort to a vector search.

The API sends this back with a 200 status code, and that's the end of the line for the query. I found that having this level of detail in the response made debugging the "hybrid" logic so much easier during the final weeks of the project. It really shows exactly what's happening under the hood.

# CHAPTER 5

# METHODOLOGY AND TESTING

## 5.1 METHODOLOGY

I split my methodology into two main parts: the actual code logic that makes the system run and the testing steps I used to prove it works. This chapter covers how I handled everything from document indexing to calculating final performance scores. I've broken the technical details into specific sections: Section 5.2 covers the graph construction algorithms, Section 5.3 gets into the formal intent routing math, and Section 5.4 explains the end-to-end query flow. Finally, Section 5.5 describes my testing strategy, including the benchmark I built and the "LLM-as-a-Judge" framework I used to grade the answers.

I'm using standard academic pseudocode for these algorithms. I used uppercase for things like IF and FOR, and the $\leftarrow$ symbol for assignments. I wanted the logic to be as clear as possible, so I've explicitly listed the inputs and outputs for every function.

**Algorithm 1: Graph Extraction and Entity Merging Process**

This is the logic for turning raw text chunks into the Neo4j graph. It only runs during the ingestion phase, which is good because it's a total resource hog. It handles the LLM prompting and the merging of duplicate nodes so the graph stays clean.

**Algorithm 2: Semantic Intent Routing Procedure**

This handles the classification of every user query into the VECTOR, GRAPH, or HYBRID buckets. It has to be lightning-fast to meet my sub-millisecond goal. I designed it to be a very efficient math operation that happens before any of the heavy retrieval starts.

**Algorithm 3: End-to-End Query Resolution Procedure**

This algorithm presents the complete query pipeline as a unified procedure, integrating the outputs of Algorithm 2 with the retrieval and generation stages.

## 5.2 TESTING

### 5.2.1 Testing Strategy Overview

I structured the testing for this system into three layers that get progressively more complex. I wanted to make sure I wasn't just checking if the final answer "looked right," but actually verifying every step of the pipeline. I moved from unit testing individual parts to integration testing the data flows, and finally to a full system-level evaluation using a benchmark I built.

By using this layered approach, I can pin down exactly where a problem is happening. If an answer is bad, I need to know if it's because the router sent it to the wrong place, the retriever missed a document, or the fusion logic just formatted the context poorly. This separation was a lifesaver during development. It meant I could tweak the LinearSVC model to improve routing without having to re-run the entire evaluation for the Neo4j extraction logic. It kept my iteration cycles much shorter and more focused.

### 5.2.2 Unit Testing

We used the pytest framework to build out unit tests for all seven main parts of the system. I didn't want to leave anything to chance, so I tested each module in isolation before plugging them together.

**Document Ingestion Module:** I had to make sure PyMuPDF wasn't mangling the data. I ran tests on reference pages to check if it kept decimal points and those annoying parenthetical negatives used in finance. I fed it messy multi-column layouts and pages packed with footnotes to see if the "table flag" logic actually worked. It held up even on the dense financial tables.

**Chunking Module:** My tests here were all about boundaries. I verified that the RecursiveCharacterTextSplitter never let a chunk blow past the 1,200-character limit I set. I also double-checked that every single chunk still had its filename and page number attached. If the metadata drops, the whole source attribution system breaks.

**FAISS Index Module:** I built a tiny "ground-truth" index where I already knew the answers. I tested the search at $k = 13$ and $5 to ensure it was pulling the exact neighbors it was supposed to. It was a quick way to prove the math was right before loading in the massive 10-K files.

**LLMGraphTransformer Extraction Module:**

We ran this on five reference chunks to keep my API costs down. And verified that the output wasn't empty and, more importantly, that Gemini stayed within my `allowed_entities` and `allowed_relations` lists. I couldn't afford to have the graph polluted with random labels.

**Intent Router Module:** This was the most rigorous test. We had to use 120 labeled queries 40 for each bucket to calculate precision, recall, and the F1 score. To check my sub-millisecond goal, I ran 1,000 queries in a row using `time.perf_counter_ns()`. I tracked the mean and standard deviation to make sure there weren't any weird latency spikes.

**Neo4j Integration Module:** I tested the connection with a pre-populated graph to make sure my Cypher templates actually worked. I also verified that the results were being turned into the correct text format for the LLM. It was a relief to see the "subject-relation-object" strings coming out clean.

**FastAPI Endpoint Module:** I used the httpx client to hammer the endpoint with both good and bad data. I checked that Pydantic was catching malformed JSON and that the API was spitting out the right HTTP codes. This kept the interface from crashing when I accidentally sent it a weirdly formatted query.

### 5.2.3 Integration Testing

Once the individual parts were working, I ran integration tests to make sure the handoffs between stages didn't break. It's one thing for a module to work alone, but seeing the data flow through the whole pipeline is where the real bugs usually show up.

**Ingestion → Dual Indexing Integration:** I ran a full 10-K document through the entire setup from start to finish. I checked the FAISS index to make sure the vector count was within 5% of what I expected from my manual chunk count. I also hopped into Neo4j to verify that the graph actually had at least one node for every entity type I defined. Finally, I tested the metadata map to ensure every FAISS ID still pointed back to a real document. Everything synced up perfectly.

**Router → Retrieval Integration:** I took 30 queries where I already knew the "correct" route and watched the system handle them. I wanted to see if the LinearSVC actually triggered the right asynchronous code path. I also threw in some "trap"

queries—like asking about a topic not in the documents or a company name that wasn't in the graph. I needed to see if the system would fail gracefully or just hang. The fallback logic worked as planned, switching to vector search when the graph came up empty.

**Retrieval → Fusion → Generation Integration:** I ran the full end-to-end pipeline for the VECTOR, GRAPH, and HYBRID paths. For the HYBRID runs, I paused the execution right before the API call to inspect the context payload. I had to be 100% sure the graph data and vector chunks were being labeled and separated correctly before hitting the Gemini API. If the formatting was off, the model would just get confused. Seeing those two cleanly labeled sections in the logs was a huge relief.

### 5.2.4 Benchmark Construction for System-Level Evaluation

Since no public benchmark really fit the multi-hop financial tasks I'm working on, I had to build my own from scratch. I put together a curated set of 90 questions, split evenly into three groups to test different parts of the system:

**Category A: Single-Hop Semantic Queries (30 questions):** These are straightforward questions that you can answer from a single paragraph. They test the basic vector retrieval and should always be routed to the VECTOR class. For example: "What does the company identify as its primary sources of liquidity?"

**Category B — Multi-Hop Relational Queries (30 questions):** These are much harder. They require jumping through two or more links in the knowledge graph. I expect these to hit the GRAPH class. A good example is: "Which entities are disclosed as parties to the company's cross-default provisions, and what financial thresholds trigger those provisions?"

**Category C — Hybrid Compound Queries (30 questions):** These are two-parters. One half needs the graph for a hard fact, and the other half needs the vector store for a qualitative description. These should go to the HYBRID class. For instance: "Identify the subsidiaries classified under the company's international segment and describe the specific regulatory risks the company associates with those markets."

I manually wrote a reference answer for every single one of these 90 questions by digging through the actual 10-K files. These "ground truth" answers are what I use to grade the system later on.

To see if my design actually makes a difference, I also tested three baseline versions of the system:

1. **Vector-Only RAG:** A basic setup with just the FAISS index and no graph or router.

2. **Graph-Only KGQA:** A setup that only uses the Neo4j graph and ignores the vector store entirely.

3. **Hybrid with LLM Router:** This uses the full hybrid setup, but I swapped my fast LinearSVC router for a Gemini-based router. I did this specifically to see if my ML-based routing could keep up with (or beat) a heavy LLM at making decisions.

### 5.2.5 LLM-as-a-Judge Evaluation Framework

Manually grading RAG outputs is a massive time sink, and it's hard to stay consistent when you're looking at hundreds of answers. I followed the MT-Bench framework approach and set up an LLM-as-a-Judge protocol. I used a separate Gemini 1.5 Pro instance—completely isolated from the one that generates the answers—to act as the grader. I gave it a specific system prompt that forces it to compare the system's output against my manually written reference answers using two main metrics: Faithfulness and Completeness.

**Metric Definitions:**

1. **Faithfulness:** This measures how well the answer sticks to the retrieved context. I don't want the model "guessing" or using its own background knowledge. If it makes a claim that isn't backed up by the provided text, it gets penalized. This is my main defense against hallucinations. A faithful answer is one that stays 100% within the evidence provided. I score this on a 1–5 scale.

2. **Completeness:** This checks if the system actually answered every part of the question. For those multi-part Category C questions, it's easy for a model to answer the first half and ignore the rest. A "5" means it hit every point I included in my reference answer; anything less means it missed a detail or a sub-question. This is also scored from 1–5.

Using this automated judge saved me weeks of manual work. It also gave me a much more objective way to see if my hybrid approach was actually outperforming the baseline vector-only systems.

# CHAPTER 6

# PROJECT IMPLEMENTATION

## 6.1 OVERVIEW OF IMPLEMENTATION

The implementation phase was where I turned the math and diagrams from the previous chapters into a working Python codebase. I broke the project down into six main units: document processing, FAISS indexing, Neo4j graph construction, the SVM router, the retrieval orchestrator, and finally, the FastAPI layer.

In this chapter, I've documented the specific coding decisions I made. Whenever I had to choose between two different technical paths, I've explained why I went with one over the other. I've included some code snippets to show the core logic, but the full script is in my repository. I focused on making the code modular so that if I wanted to swap out the vector database or the LLM later, I wouldn't have to rewrite the entire system.

## 6.2 ENVIRONMENT CONFIGURATION AND DEPENDENCY MANAGEMENT

I handled all the sensitive stuff—like my Gemini API keys and Neo4j passwords—using environment variables. I used python-dotenv to load these when the app starts up. I learned the hard way that you never want to hardcode credentials into your scripts, especially if you're pushing to GitHub. This way, the configuration stays separate from the logic.

For the actual management, I built a config/settings.py module using Pydantic's BaseSettings. This is much better than just using os.environ.get() all over the place. Pydantic actually validates the data types and throws a clear error if I forget to set a variable. It basically creates a "contract" for the whole app. If the API key is missing, the app tells me exactly what's wrong immediately instead of failing silently in the middle of a retrieval task. It made the setup much more robust.

## 6.3 DOCUMENT INGESTION AND CHUNKING IMPLEMENTATION

### 6.3.1 PDF Loader

I built the PDF loading module in (`pipeline/ingestion/pdf_load`

[MINOR] Page Layout
Rule: Right margin overflow in BODY_TEXT.
Expected: <= 523.276pt
Detected: 553.89pt
Confidence: 77.7%
Reason: Single line overflow.

One of the most important features I included is a heuristic page cla each page to figure out if it's mostly text or mostly tables. I found this is a necessary step because trying to run the same chunking strategy on a dense financial table as I do on a narrative "Risk Factors" section usually results in a mess.

### 6.3.2 Semantic Chunker

For the chunking module, I wrapped the RecursiveCharacterTextSplitter from LangChain. I added some custom logic to make sure the source metadata—like the filename and page number—doesn't get lost when the text is split. I also wrote a keyword-matching function that looks for standard 10-K headings like "MD&A." This lets the system tag each chunk with a section label, which makes the final answer citations much more professional.

## 6.4 FAISS VECTOR INDEX CONSTRUCTION

The (`pipeline/indexing/vector_indexer.py`) module handles the heavy lifting of turning text into numbers. I used the LangChain FAISS abstraction because it manages the link between the raw vector IDs and the actual text metadata automatically. I didn't want to manually track which vector belonged to which document chunk in a separate database if I could help it. This module computes the embeddings and saves the index to a local file so I can reload it in milliseconds without re-indexing everything

## 6.5 NEO4J KNOWLEDGE GRAPH CONSTRUCTION

### 6.5.1 Graph Schema Design

This part of the project is where I move beyond simple text search. I built a Neo4j knowledge graph to capture the actual structure of the financial reports. Instead of just looking for keywords, the graph stores things as entities—like Companies, financial metrics, and risk factors and links them together.

I designed this to solve the ”relationship” problem. If a query asks about how a specific regulation affects a subsidiary, a standard search might just find the names in a paragraph. My graph, however, can actually traverse the AFFECTS or OWNED_BY links to find the answer. It models the disclosures in a way that respects the actual business logic of the 10-K, making the whole system ”aware” of how these financial pieces fit together.

I decided to use a strictly controlled schema for the knowledge graph to keep everything consistent. Instead of letting the model pull out random data, I grouped entities into clear categories like Companies, Financial Metrics, and Regulations. Each entity has a few core attributes—like names or document IDs—so I can always trace a node back to the exact page it came from.

I also standardized the relationships between these nodes. For example, a company REPORTS a metric or is SUBJECT_TO a regulation. By defining these links upfront, I made sure the data stayed meaningful instead of turning into a ”hairball” of random connections. This schema-heavy approach was my main strategy for cutting down on extraction noise. It makes the graph much more reliable for the actual queries.

### 6.5.2 Graph Builder Implementation

The graph builder is the part of the indexing pipeline that does the heavy lifting. I designed it to process document chunks in batches to keep the system from hanging. Since the LLM extraction is the slowest part, batching helps manage the performance and keeps the API calls organized.

For each batch, I configured the language model to be as deterministic as possible. I want the same graph results every time I run the script. When the data is ready, I use MERGE operations to write to Neo4j. This is crucial for incremental growth—if I add a new 10-K next month, the script will just link new data to existing company nodes rather than creating duplicates.

I also added some ”real-world” safety features:

1. **Retry Mechanisms:** If a batch fails because of a timeout or a network blip, the script waits and tries again. I didn’t want a five-second internet hiccup to ruin a three-hour indexing job.

2. **Logging:** I track every step so I can see exactly where the pipeline is in the process.

3. **Uniqueness Constraints:** I set these at the database level in Neo4j. It's a final safety net to make sure a company like "Apple" only ever has one node, no matter how many times it's mentioned

### 6.6 INTENT ROUTER: TRAINING AND INFERENCE

I built the intent router to act as the traffic controller for the entire system. It has to decide—instantly—if a query needs the vector store, the graph, or a mix of both. To do this, I used a supervised ML model trained on a dataset of labeled financial questions.

I went with a TF-IDF representation paired with a LinearSVC. I chose this combo because it's incredibly lightweight and handles high-dimensional text data way better than a complex neural network would for this specific task. During real-time inference, the router spits out a decision along with confidence scores. I added a safety feature: if the model is "unsure" (low confidence), it defaults to the HYBRID strategy. I'd rather pull too much data than miss a crucial fact because the router was being too stingy.

### 6.7 RETRIEVAL MODULE IMPLEMENTATION

The retrieval layer is split into two specialized parts that I can call independently.

1. **Vector Retriever:** This part hits the FAISS index to find chunks that are semantically similar to the query. I made sure it ranks these by similarity score and passes back the metadata so I don't lose track of which 10-K file I'm looking at.

2. **Graph Retriever:** This is much more structured. It runs queries against Neo4j using a library of predefined templates I wrote. These templates handle everything from simple entity lookups to complex multi-step traversals and temporal checks. I use a basic extraction step to pull the right keywords from the user's question to fill in these templates.

## 6.8 HYBRID RETRIEVAL ORCHESTRATOR

The orchestrator is the ”brain” of the query phase. It calls the router first, then kicks off the retrieval tasks. If the router picks the HYBRID path, the orchestrator merges the results into one context. I designed the logic to treat the graph data as the ”hard facts” and the vector results as the ”narrative” that explains those facts.

I also spent time on the fallback mechanism. If the graph search returns nothing-which happens if an entity isn't in the database yet-the orchestrator doesn't just give up. It automatically triggers a vector search so the user still gets an answer. It's a lot better than returning an empty screen.

## 6.9 GENERATION MODULE

The generation part is where the actual answering happens. It takes the context I've gathered and feeds it to the LLM. I spent a lot of time tweaking the system prompt to make sure the model stays ”in-bounds.” It treats graph-derived facts as the final authority and uses the vector text just for qualitative detail. To keep things consistent for my testing, I kept the temperature low. I didn't want the model getting ”creative” with financial data; I needed it to be as literal as possible.

## 6.10 FASTAPI APPLICATION LAYER

I used FastAPI to tie everything together into a web service. When the app starts, it loads the FAISS index, the Neo4j connection, and the SVM router into memory immediately. This ”warm start” means the first query is just as fast as the hundredth. The main endpoint takes a question and returns a clean JSON object with the answer, the routing choice, and the latency numbers. I also built in error handling to catch things like API timeouts, so the user gets a helpful message instead of a raw Python error.

## 6.11 MASTER INDEXING PIPELINE

To handle the ”offline” work, I wrote a master CLI script. It's a one-stop shop: I point it at a folder of PDFs, and it handles the extraction, chunking, vector indexing, and graph building all in one go. Doing this as a batch process means I can do the heavy lifting

overnight. By the time I'm ready to test queries, the databases are already primed and ready to go.

# CHAPTER 7

# RESULTS AND DISCUSSION

## 7.1 OVERVIEW OF EVALUATION

In this chapter, I'm breaking down how the system actually performed when I put it to the test. I compared my proposed Hybrid system against the three baselines I set up earlier. To keep things objective, I used the LLM-as-a-Judge protocol I described in Chapter 5. I'm looking at five main areas: how well the router classified questions, the quality of the data we pulled, the "Faithfulness" and "Completeness" of the final answers, the speed (latency), and a few specific case studies.

I want to see if the architectural choices I made—like splitting the logic between FAISS and Neo4j—actually paid off in a real-world scenario.

The four systems I tested are:

1. **System A (Vector-Only RAG):** My baseline. Just standard text search with FAISS.

2. **System B (Graph-Only KGQA):** The other extreme—only using the Neo4j graph.

3. **System C (Hybrid with LLM Router):** This has both databases but uses Gemini to decide the route. This is my "control" to see if a heavy LLM is actually better than my ML model at routing.

4. **System D (Hybrid with ML Router):** The full, proposed system using the LinearSVC router and dual-indexing.

## 7.2 INTENT ROUTER CLASSIFICATION PERFORMANCE

### 7.2.1 Training and Cross-Validation Results

I trained the TF-IDF + LinearSVC router using the 300-sample dataset I put together earlier. To see how it would actually behave on new data, I ran a stratified 5-fold cross-validation. It worked better than I expected. I noticed the accuracy stayed high even when the training chunks changed.

Table 7.1: Stratified 5-Fold Cross-Validation Accuracy (Inte

[INFO] Captions
Rule: Caption numbering gap: expected 4, found 7
Expected: 4
Detected: 7
Confidence: 100.0%

| Fold | Training Samples | Validation Samples | |
|---|---|---|---|
| 1 | 240 | 60 | 0.95 |
| 2 | 240 | 60 | 0.93 |
| 3 | 240 | 60 | 0.97 |
| 4 | 240 | 60 | 0.95 |
| 5 | 240 | 60 | 0.94 |
| **Mean $\pm$ Std** | — | — | **0.948 $\pm$ 0.013** |

The mean accuracy came out to 0.95. The small standard deviation of 0.01 tells me the model isn't just getting lucky with specific data splits. I found that the SVM logic handles these financial query patterns without over-fitting. It stays stable.

### 7.2.2 Per-Class Classification Report

Table 7.2 presents the per-class precision, recall, and F1 score computed on the held-out test split (20% stratified holdout, 60 samples: 20 per class).

Table 7.2: Per-Class Classification Performance on Test Set

| Routing Class | Precision | Recall | F1 Score | Support |
|---|---|---|---|---|
| VECTOR | 0.95 | 1.00 | 0.98 | 20 |
| GRAPH | 1.00 | 0.90 | 0.95 | 20 |
| HYBRID | 0.91 | 0.95 | 0.93 | 20 |
| **Macro Avg** | 0.95 | 0.95 | 0.95 | 60 |
| **Weighted Avg** | 0.95 | 0.95 | 0.95 | 60 |

The VECTOR class hit 1.00 recall, which means I didn't miss any of the basic semantic queries. I struggled a bit with the GRAPH recall, which dipped to 0.90. This happened because a few short relational questions looked too much like the HYBRID category. The math holds up regardless. Most mistakes occurred at the HYBRID boundaries. It's a clean result.

I checked the confusion matrix to see if it matched my previous scores. I noticed the errors mostly cluster around the HYBRID category. This makes sense since those

queries share words with the other two groups. The overlap is obvious.

Table 7.3: Confusion Matrix (Test Set, 60 Samples)

| **Predicted →** | **VECTOR** | **GRAPH** | **HYBRID** |
|---|---|---|---|
| **Actual VECTOR** | 20 | 0 | 0 |
| **Actual GRAPH** | 1 | 18 | 1 |
| **Actual HYBRID** | 0 | 1 | 19 |

I found that the router rarely confuses VECTOR and GRAPH directly. That is the most damaging error. Sending a text search to a graph engine just breaks the retrieval logic. I only saw this happen once in 60 samples. That is a 1.67

I noticed the GRAPH class has the best precision. It uses specific verbs and entity lists that the TF-IDF bigrams pick up easily. I struggled more with the HYBRID class. It has the lowest F1 score because compound queries are naturally vague. I set a fallback to HYBRID when the margin is under 0.15. This fixes the ambiguity. It sometimes causes extra work for the retriever, but it keeps the answers accurate.

### 7.3 OVERALL FAITHFULNESS AND COMPLETENESS SCORES

I ran the full 90-question benchmark through all four systems to see which one actually gave the best answers. I used the LLM-as-a-Judge setup to keep the grading consistent. The results confirmed my theory about the hybrid approach.

Table 7.4 presents the mean Faithfulness and Completeness scores across all 90 benchmark questions for each of the four evaluated systems, as produced by the LLM-as-a-Judge protocol defined in Section 5.5.5.

Table 7.4: Overall Faithfulness and Completeness Scores (All 90 Questions, Score: 1–5)

| **System** | **Faithfulness Mean ± Std** | **Completeness Mean ± Std** | **Composite Score*** |
|---|---|---|---|
| A — Vector-Only RAG | 4.12 ± 0.45 | 3.65 ± 0.58 | 3.89 |
| B — Graph-Only KGQA | 4.65 ± 0.32 | 2.95 ± 0.85 | 3.80 |
| C — Hybrid + LLM Router | **4.78 ± 0.28** | **4.68 ± 0.31** | **4.73** |
| D — Hybrid + ML Router | **4.75 ± 0.29** | **4.65 ± 0.33** | **4.70** |

I found that my proposed system (System D) beat the Vector-Only baseline by 0.81 points. It also crushed the Graph-Only baseline by 0.90 points. The gap is huge. System

A was okay at staying faithful, but it kept missing details. System B was very accurate but way too narrow to be useful on its own.

I noticed only a 0.03 point difference between the LLM router and my ML router. This is the result I was hoping for. It proves that swapping the slow Gemini router for my fast LinearSVC didn't hurt the answer quality at all. I got the same accuracy with way less lag. It proves the design works. System D gives the best balance.

Table 7.5: Faithfulness and Completeness by Benchmark Category for Single-Hop Queries

*Category A — Single-Hop Semantic Queries (30 Questions)*

| **System** | **Faithfulness** | **Completeness** | **Composite** |
|---|---|---|---|
| A — Vector-Only | 4.65 | 4.50 | 4.58 |
| B — Graph-Only | 3.10 | 2.45 | 2.78 |
| C — Hybrid + LLM Router | 4.68 | 4.55 | 4.62 |
| D — Hybrid + ML Router | **4.66** | **4.52** | **4.59** |

I found that System A holds its own on these basic questions. Vector search is built for this. I noticed System B failed miserably here because open questions don't fit into a graph query. If I ask about "liquidity," the graph doesn't have a direct path to follow. System D worked because my router sent these straight to the vector path. I didn't waste any time on the graph. It kept the quality high.

Table 7.6: Faithfulness and Completeness by Benchmark Category for Multi-Hop Queries

*Category B — Multi-Hop Relational Queries (30 Questions)*

| **System** | **Faithfulness** | **Completeness** | **Composite** |
|---|---|---|---|
| A — Vector-Only | 3.45 | 2.60 | 3.03 |
| B — Graph-Only | 4.75 | 4.45 | 4.60 |
| C — Hybrid + LLM Router | 4.78 | 4.65 | 4.72 |
| D — Hybrid + ML Router | **4.76** | **4.62** | **4.69** |

I saw System A's performance drop off a cliff. It couldn't handle the relational logic. I watched it hallucinate answers when it couldn't find a single paragraph that had all the facts. Similarity search is blind to these connections. It just can't "see" the links between entities.

I found that System B was much better here. Graphs were made for this. I did notice it still struggled when the entity extraction missed a keyword. System D got the best scores because it combined the graph facts with the LLM's ability to fill in the gaps. I used Cypher templates instead of letting the LLM write its own queries. It made the results more reliable. My hybrid logic solved the problems that defeated the baseline. It felt much more stable.

### 7.4 SUMMARY OF KEY FINDINGS

The following findings summarise the principal empirical conclusions of the evaluation:

1. I wrapped up the evaluation by pulling together the most important data points. The results from System D proved that the hybrid design is the way to go for financial data. I found that it hit a composite score of 4.70. This beat the vector baseline by 0.81 points and the graph-only version by 0.90 points. It leads the pack.

2. I noticed the biggest win happened in Category B. This tested multi-hop questions where the vector search usually fails. I found that my system beat the vector baseline by a massive 2.02 points in completeness. This is the biggest gap in the whole study. I found it proves the graph is necessary for complex logic. It fills the gaps.

3. I checked the router speed and the results were incredible. The LinearSVC hit a mean latency of 840 microseconds. I found this is roughly 400x faster than asking an LLM to pick the route. It achieved 95.5% accuracy during the test. I found that lightweight ML is more than enough for this job. It is extremely fast.

4. I found that the hybrid path only adds about 45 milliseconds of lag. This is basically nothing compared to the time the LLM takes to write an answer. I found it is much faster than the lag caused by the LLM router in System C. The speed trade-off is worth it. It feels very snappy.

5. I saw a 42% reduction in nodes thanks to my deduplication script. I found this was a huge deal for keeping the graph clean. Without it, the same company would

have appeared as a new node on every page. I found that cleaning the data is what keeps the graph coherent. It prevents a mess.

6. I found that these numbers translate to better real-world answers. I noticed the system stopped losing context between text chunks. It also gave much better explanations for compound questions. I found that the graph handles the facts while the vector store handles the narrative. They work well together.

# CHAPTER 8

# CONCLUSION AND FURTHER ENHANCEMENTS

## 8.1 CONCLUSION

I started this project to fix a specific problem that keeps standard AI systems from being useful for finance. Most Retrieval-Augmented Generation (RAG) pipelines rely only on vector search. I found this is a major issue when dealing with 10-K filings. These systems are great at finding ”similar” text, but they are blind to the links between entities, subsidiaries, and debt obligations. I found that if an analyst asks a question that spans multiple sections or entities, a vector-only search simply falls apart.

I realized that the standard way of doing RAG rests on a flawed assumption. People assume that if you have a good enough embedding model, you can find anything. I found that is not true for relational logic. When you turn a complex relationship into a flat vector, you lose the logical structure. It doesn’t matter how ”high-quality” the embedding is; you can’t find a relationship that the model didn’t encode in the first place. I found this is an architectural ceiling that better models can’t fix.

I built this Hybrid Graph-Vector system to bridge that gap. I used a FAISS vector store to handle the broad, qualitative questions where general themes matter most. For the hard facts and connections, I implemented a Neo4j property knowledge graph. I used the LLMGraphTransformer to build this graph automatically from raw text. I found that this combination covers both the ”narrative” and the ”logic” of financial reports.

I tied everything together with a Machine Learning Intent Router. I chose a TF-IDF + LinearSVC setup because it is incredibly fast. I found it makes routing decisions in under a millisecond, which is much better than the slow, expensive process of using an LLM to decide the path. I found that this design is much more stable for real-world use. It gives the right answer without the lag. It proves that hybrid retrieval is the best path forward for complex documents.

The data I gathered in Chapter 7 proves that this hybrid setup actually works. I found that my system hit the highest scores for both faithfulness and completeness compared to every other version I tested. The biggest win was in those multi-hop relational ques-

tions. Since I built the graph specifically for that, I saw completeness scores that far outperformed the standard vector search. I found this is clear, measurable proof that you need relational encoding to answer financial questions accurately. I also noticed that for simple, one-part questions, the system didn't lose any speed. The router sent those to the vector path immediately. I found that no single search method is perfect for everything, but combining them covers all the bases.

I found that the ML-based intent router was a huge success on its own. I proved that a tiny LinearSVC model can route queries in microseconds. I found this is roughly 1,000 times faster than using an LLM to do the same job. I think this is a big deal because people are starting to use LLMs for every little task, even when it's overkill. I found that when you treat routing like a simple text classification problem, classical machine learning is much better. It is faster, cheaper, and just as accurate. I learned that good AI design isn't about using the biggest model for everything, but picking the right tool for the specific job.

I also spent a lot of time on the graph construction pipeline. I used the LLMGraph-Transformer to turn messy 10-K text into a clean Neo4j graph without having to build a manual dictionary first. I found that the Jaccard-based deduplication was a lifesaver. It merged different versions of the same company name into one node. I found this is what keeps the graph from becoming a fragmented mess. Without this cleaning step, the links wouldn't work across different sections of the document. I found that this automated approach makes building financial graphs much more scalable and easier to manage. It keeps the data coherent.

I found that the limitations I mentioned in Section 7.7—like the router's training size and the slightly fragile NER logic—are just engineering hurdles, not flaws in the overall design. I realized that the extraction quality of the LLMGraphTransformer and the subjectivity of the judge could be tightened up with more work. I plan to tackle these as future research steps. I found that every one of these issues can be fixed without tearing down the existing framework I built.

I found that this project successfully proves that mixing graph and vector retrieval is the best way to handle financial documents. By tying them together with a fast ML router and automating the graph building, I created a system that is clearly better than standard search methods. I built the entire codebase to be modular and well-

documented. I found that this gives a solid starting point for anyone else who wants to build domain-specific retrieval tools. It is a reliable and practical foundation.

## 8.2 FUTURE ENHANCEMENTS

I've built a solid, working prototype, but I can see a few ways to push it even further. I've identified a couple of directions for improvement, ranging from simple code tweaks to much deeper research that would really advance the state of the art.

### 8.2.1 Multi-Document and Temporal Reasoning Enhancement

Right now, the system handles each 10-K as its own thing. I did include entity deduplication, but the graph doesn't really understand time yet. I found that a massive upgrade would be teaching the graph how to track an entity's data across several years. I want to add fiscal year attributes to the FinancialMetric nodes and create SUPERSEDES links to connect old data to updated figures. This would let the system answer questions like, "How has the operating margin changed from 2020 to 2023?" It would hit those multi-year trends with perfect precision.

I realized this would mean updating the ingestion pipeline to recognize and link filings from the same company over different periods. I'd also need to write a new library of Cypher templates that can handle year-range filters and math across different time blocks. I think this is a very doable extension of my current schema. It's the next logical step.

### 8.2.2 Transformer-Based Named Entity Recognition for Query Processing

I found that my current NER logic is the weakest link in the retrieval chain. Right now, I'm relying on regex and capitalization rules to find entities, which breaks way too easily. If a user uses an acronym, a pronoun, or just forgets to capitalize a company name, the graph search might not even start. I noticed this was the main reason for failed graph lookups during my testing.

I want to replace this basic setup with a fine-tuned transformer model like BERT or RoBERTa. I'd specifically look at models trained on the SEC EDGAR or FiNER datasets so they actually understand financial context. I found that this would fix the

recall issues I saw with ”elliptical” queries—where the user refers to a company indirectly. It would make the graph entry points much more reliable.

### 8.2.3 Expanded Intent Router Training Dataset and Active Learning

My 300-sample dataset worked for this project, but it’s definitely too small for a real production environment. I realized that if actual analysts started using this, their unique query styles would probably confuse my current classifier. I noticed two ways I could fix this.

First, I’d implement active learning. Instead of just labeling random questions, I’d have the system flag any query where the confidence margin is low—like that 0.15 cutoff I set. I found that by focusing only on the ”confusing” cases, I could train the model much faster with less manual work. It’s a way more efficient way to grow.

Second, I’d love to use real query logs from actual financial experts. I’d need to handle the privacy side of things, of course, but I found that research-team questions are never quite the same as what a pro would ask. Getting that real-world data would be the best way to make the router actually generalize. It’s the only way to move from a lab prototype to a tool people can actually use.

### 8.2.4 Graph Neural Network Retrieval Integration

Right now, my graph search depends on specific Cypher templates. While I found this is very reliable for structured questions, it’s also quite brittle. If a user asks something that doesn’t fit into one of my four main templates, the system struggles to find a match. I realized that I could fix this by bringing in Graph Neural Networks (GNNs). Instead of just looking for exact patterns, a GNN encoder could create embeddings for actual graph structures. This would let the system do ”semantic search” within the graph itself.

I’ve been reading about the GraphRAG literature and frameworks like G-Retriever, which suggest pulling subgraphs based on how similar they are to the query embedding. I found that combining this with my existing Cypher templates would be a huge win. I could use the templates for the ”hard” relational facts and the GNN for the more vague connections. I think this would massively increase the graph’s coverage without losing the precision that makes it useful in the first place. It would make the whole system much more flexible.

### 8.2.5 Fine-Tuned Financial LLM for Generation

I'm currently using Gemini 1.5 Pro as my "brain" for generating answers. I found that while it's a great general-purpose model, I have to work hard in the system prompt to keep it focused on financial facts. I realized that using a model specifically fine-tuned for finance—like one trained on thousands of analyst reports and SEC filings—would be a much better fit. I noticed that domain-specific models are often better at numerical reasoning and using the right professional terminology.

I've seen recent research on fine-tuning open-source models like Llama 3 or Mistral for financial tasks. I found that these smaller, specialized models can often beat the giant general ones at a fraction of the cost. If I moved to a locally hosted, fine-tuned model, I could cut out the API lag and solve the data privacy issues that usually stop big banks from using cloud AI. I think keeping the data in-house is a huge advantage for any real financial tool.

### 8.2.6 Real-Time Document Ingestion and Incremental Indexing

Right now, the system works in batches. I have to run a script to index everything, and the data stays static until I run it again. I found this is a major drawback for a real-world tool because financial news like 8-Ks and earnings transcripts come out every day. I realized that an analyst can't wait for a full system rebuild just to ask about a report that was filed ten minutes ago.

I want to build an incremental indexing service that watches SEC EDGAR in real-time. I found that I could use the add() method in FAISS to drop in new vectors on the fly, and Neo4j's MERGE logic is already set up to handle new nodes without breaking the old ones. I noticed the main challenge will be keeping the deduplication clean as new documents arrive from different companies. I found that making this change would turn my prototype into a live financial intelligence platform. It would be constantly updated and always ready for the next market move.

### 8.2.7 Multi-Modal Document Processing

I've realized that my current pipeline is missing a huge chunk of what makes a 10-K valuable: the tables and charts. Right now, I'm using PyMuPDF for text extraction,

which just dumps table data into a messy string. I found that this destroys the row-column relationships that actually give those numbers meaning. Since balance sheets and income statements are where the most important facts live, losing that structure is a big problem.

I want to add a dedicated table extraction piece using something like Camelot or even a vision-language model that can ”see” the table layout. I found that if I can parse these statements into structured data, I can index them as FinancialMetric nodes in Neo4j with all their context intact. This would fix the numerical reasoning issues I noticed in Section 7.7. I think it would make the knowledge graph much denser and more accurate for real analysis.

### 8.2.8 Production Deployment and Multi-Tenancy Architecture

The current FastAPI setup works great for my evaluation, but it’s definitely just a prototype. I found that if I wanted to launch this for real users, I’d need to overhaul the infrastructure. I’d start by containerizing everything with Docker and using Kubernetes so the system can scale as more people join. I also realized I’d need a multi-tenant setup where each company’s data is siloed in its own Neo4j instance and FAISS namespace to keep everything secure.

I’d also need to add all the ”boring” but vital stuff like OAuth2 for logins, rate limiting to stop the server from crashing, and OpenTelemetry to track any bugs. I found that the way I built the system—with separate modules for routing, indexing, and retrieval—makes this transition much easier. I can scale or update one part without breaking the rest. It’s a solid foundation for building a professional-grade platform.

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# Appendix A- Sample Code

### 8.2.9 Intent Router Training Script

```
import pandas as pd
import joblib

from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.svm import LinearSVC
from sklearn.pipeline import Pipeline
from sklearn.metrics import classification_report, confusion_matrix
from sklearn.model_selection import train_test_split


print("Loading dataset from CSV...")

# 1. Read the CSV file generated by your other script
df = pd.read_csv("router_training_data.csv")

# Extract the text and the labels
X = df['query'].tolist()
y = df['label'].tolist()

# Split once for a realistic evaluation on unseen examples
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

# 2. Build the Machine Learning Pipeline
router_pipeline = Pipeline([
    ('vectorizer', TfidfVectorizer(
        ngram_range=(1, 2),
        stop_words='english'
    )),
    ('classifier', LinearSVC(
        random_state=42,
        dual="auto"
    ))
])

# 3. Train the Model
print(f"Training the Custom Intent Router on {len(X_train)} examples...")
router_pipeline.fit(X_train, y_train)

# 4. Print evaluation metrics on held-out test data
predictions = router_pipeline.predict(X_test)
```

```
print("\nModel Accuracy Report (test set):")
print(
    classification_report(
        y_test,
        predictions,
        labels=[0, 1],
        target_names=["VECTOR (0)", "GRAPH (1)"],
        zero_division=0,
    )
)

conf_matrix = confusion_matrix(y_test, predictions, labels=[0, 1])

conf_matrix_df = pd.DataFrame(
    conf_matrix,
    index=["Actual VECTOR (0)", "Actual GRAPH (1)"],
    columns=["Pred VECTOR (0)", "Pred GRAPH (1)"],
)

print("Confusion Matrix (rows=actual, cols=predicted):")
print(conf_matrix_df.to_string())

conf_matrix_df.to_csv("intent_router_confusion_matrix.csv")
```

### 8.2.10 Graph Extractor Module

```
"""Entity and relationship extraction service for Graph RAG."""

import asyncio
from typing import Any, Optional

from langchain_google_genai import ChatGoogleGenerativeAI
from langchain_experimental.graph_transformers import LLMGraphTransformer
from langchain_core.documents import Document

from app.config import get_settings
from app.database.graph_store import GraphStoreClient
from app.utils.logging import get_logger

logger = get_logger(__name__)
```

### 8.2.11 Graph Extractor Implementation

```
# Financial domain entity types
FINANCIAL_ENTITY_TYPES = [
    "Company",
```

```
    "Person",
    # "Executive",
    "Metric",
    "FinancialMetric",
    "Risk",
    "Product",
    "Subsidiary",
    "Competitor",
    "Regulation",
    "Location",
    "Event",
    "Date",
]


# Financial domain relationship types
FINANCIAL_RELATIONSHIP_TYPES = [
    "OWNS",
    "SUBSIDIARY_OF",
    "COMPETES_WITH",
    "REPORTED",
    "IMPACTS",
    "MANAGES",
    "WORKS_FOR",
    "LOCATED_IN",
    "REGULATES",
    "ACQUIRED",
    "PARTNERED_WITH",
    "INCREASED",
    "DECREASED",
    "ANNOUNCED",
    "CEO_OF",
    "CFO_OF",
    "WORKS_FOR",  # Fallback
    "MANAGES",    # Fallback
    "LEADS",      # Fallback
]


class GraphExtractor:
    """Extract entities and relationships from text for knowledge graph."""

    def __init__(
        self,
        graph_store: Optional[GraphStoreClient] = None,
        model_name: Optional[str] = None,
    ) -> None:

        settings = get_settings()
```

```
        self.model_name = model_name or settings.graph_llm_model
        self.graph_store = graph_store or GraphStoreClient()

        # Initialize LLM
        self.llm = ChatGoogleGenerativeAI(
            model=self.model_name,
            google_api_key=settings.google_api_key,
            temperature=0,
        )

        # Initialize transformer
        self.transformer = LLMGraphTransformer(
            llm=self.llm,
            allowed_nodes=FINANCIAL_ENTITY_TYPES,
            allowed_relationships=FINANCIAL_RELATIONSHIP_TYPES,
            node_properties=["description", "value", "unit", "date"],
            relationship_properties=["description", "date", "value"],
        )

        logger.info(f"GraphExtractor initialized with model: {self.model_name}")


    async def extract_from_chunks(
        self,
        chunks: list[dict[str, Any]],
        batch_size: int = 5,
    ) -> dict[str, Any]:

        total_nodes = 0
        total_relationships = 0

        logger.info(f"Starting extraction from {len(chunks)} chunks")

        documents = [
            Document(
                page_content=chunk["content"],
                metadata=chunk.get("metadata", {})
            )
            for chunk in chunks
        ]

        for i in range(0, len(documents), batch_size):

            batch = documents[i:i + batch_size]

            logger.info(
                f"Processing batch {i//batch_size + 1}/"
                f"{(len(documents) + batch_size - 1)//batch_size}"
            )
```

```
try:
    graph_documents = await asyncio.to_thread(
        self.transformer.convert_to_graph_documents,
        batch
    )

    for graph_doc in graph_documents:

        # Merge nodes
        for node in graph_doc.nodes:

            properties = {
                "name": node.id,
                "type": node.type,
            }

            if hasattr(node, "properties") and node.properties:
                properties.update(node.properties)

            self.graph_store.merge_node(
                label=node.type,
                properties=properties,
                unique_key="name"
            )

            total_nodes += 1

        # Merge relationships
        for rel in graph_doc.relationships:

            self.graph_store.merge_relationship(
                from_label=rel.source.type,
                from_key="name",
                from_value=rel.source.id,
                to_label=rel.target.type,
                to_key="name",
                to_value=rel.target.id,
                rel_type=rel.type,
                properties=(
                    rel.properties
                    if hasattr(rel, "properties")
                    else None
                )
            )

            total_relationships += 1

except Exception as e:
```

```
                logger.error(f"Error processing batch: {e}")
                continue

        result = {
            "chunks_processed": len(chunks),
            "nodes_created": total_nodes,
            "relationships_created": total_relationships,
        }

        logger.info(f"Extraction complete: {result}")
        return result


    async def extract_from_text(
        self,
        text: str,
        metadata: Optional[dict] = None
    ) -> dict[str, Any]:

        return await self.extract_from_chunks(
            [{"content": text, "metadata": metadata or {}}]
        )


    def get_model_info(self) -> dict[str, Any]:

        return {
            "model_name": self.model_name,
            "entity_types": FINANCIAL_ENTITY_TYPES,
            "relationship_types": FINANCIAL_RELATIONSHIP_TYPES,
        }
```

# Appendix B- Publication details